Writing Numbers in Scientific Notation

Jonathan R. Bacolod

Scientific notation is a way of writing very large or very small numbers in a shorter form.

Why Use It?

Some numbers are too long to write easily.

$$300\,000\,000$$

$$0.000\,000\,5$$

Scientific notation makes them short.

The Form

A number in scientific notation is written as

$$a \times 10^n$$

Parts of Scientific Notation

$$\underbrace{a}_{\text{coefficient}} \times \underbrace{10}_{\text{base}}^{\overbrace{n}^{\text{exponent}}}$$

The Coefficient

a is the **coefficient**.

It must be at least 1 but less than 10.

$$1 \leq a < 10$$

Coefficient: Yes or No?

Number	Coefficient?
4.5	Yes
1	Yes
45	No
0.45	No

The Base and the Exponent

10 is the **base**.

n is the **exponent**, which is an integer.

$$6.2 \times 10^4$$

Writing in Scientific Notation

Step 1: Move the decimal point until the coefficient is at least 1 but less than 10.

Writing in Scientific Notation

Step 2: Count the number of places the decimal point moves.

Writing in Scientific Notation

Step 3: Choose the sign of the exponent.

Positive or Negative Exponent?

Original Number	**Exponent**
10 or greater	Positive
Less than 1 (not zero)	Negative

This is based on absolute value.

Remember

Large numbers usually have **positive** exponents.

Small positive numbers less than 1 have **negative** exponents.

Back to Ordinary Form

To write scientific notation in ordinary decimal form, look at the exponent.

Positive Exponent

A **positive exponent** moves the decimal point to the **right**.

$$\text{Positive} \Rightarrow \text{Right}$$

Negative Exponent

A **negative exponent** moves the decimal point to the **left**.

$$\text{Negative} \Rightarrow \text{Left}$$

Add Zeros

Add zeros when necessary to show the correct place value.

$$3 \times 10^4 = 30\,000$$

Memory Tip

Exponent	Move	Size
Positive	Right	Big
Negative	Left	Small

Examples

Example 1

Write $45\,000$ in scientific notation.

Move the decimal point 4 places to the left.

$$45\,000 = 4.5 \times 10^4$$

The number is large, so the exponent is positive.

Example 2

Write 0.0032 in scientific notation.

Move the decimal point 3 places to the right.

$$0.0032 = 3.2 \times 10^{-3}$$

The number is less than 1, so the exponent is negative.

Example 3

Write 5.1×10^6 in ordinary decimal form.

The exponent is positive, so move the decimal point 6 places to the right.

$$5.1 \times 10^6 = 5\,100\,000$$

Example 4

Write 7.4×10^{-4} in ordinary decimal form.

The exponent is negative, so move the decimal point 4 places to the left.

$$7.4 \times 10^{-4} = 0.00074$$

Practice Exercises

Practice 1

Write the number in scientific notation.

$$72\,000$$

Practice 2

Write the number in scientific notation.

$$5\,630\,000$$

Practice 3

Write the number in scientific notation.

$$408\,000\,000$$

Practice 4

Write the number in scientific notation.

$$0.00056$$

Practice 5

Write the number in scientific notation.

$$0.00000819$$

Practice 6

Write the number in ordinary decimal form.

$$6.4 \times 10^5$$

Practice 7

Write the number in ordinary decimal form.

$$2.75 \times 10^7$$

Practice 8

Write the number in ordinary decimal form.

$$8.3 \times 10^{-3}$$

Practice 9

Write the number in ordinary decimal form.

$$4.62 \times 10^{-5}$$

Practice 10

Write the number in ordinary decimal form.

$$9.01 \times 10^2$$

Activities

Activity 1

Write the number in scientific notation.

$$91\,000$$

Activity 2

Write the number in scientific notation.

$$8\,240\,000$$

Activity 3

Write the number in scientific notation.

$$637\,000\,000$$

Activity 4

Write the number in scientific notation.

$$0.00083$$

Activity 5

Write the number in scientific notation.

$$0.00000642$$

Activity 6

Write the number in ordinary decimal form.

$$7.2 \times 10^5$$

Activity 7

Write the number in ordinary decimal form.

$$3.48 \times 10^7$$

Activity 8

Write the number in ordinary decimal form.

$$6.9 \times 10^{-3}$$

Activity 9

Write the number in ordinary decimal form.

$$5.37 \times 10^{-5}$$

Activity 10

Write the number in ordinary decimal form.

$$8.04 \times 10^2$$

Thank you!